

ELITE MATHEMATICS

Pearson Edexcel International A Level

Pure 2

Adaptive Classified

Chapter 12 • Integration

Questions

Student book • questions with writing space

- 90 pages
- Past papers up to January 2026

PURE 2 | TOPIC 12

Integration

Your classified route

29

QUESTIONS

212

MARKS

4

FAMILIES

5

VARIANTS

DR ESLAM AHMED

CLASSIFIED BY IDEA

Move from recognition to method, then use the working space to complete the solution.

TOPIC 12 | FAMILY 01

Exact Area

WHAT THIS FAMILY ASKS

Recognise the exact area demand and complete the required reasoning in a clear, exam-ready order.

VARIANT ROUTE

V01 core family move

9 classified questions

78 total marks

8. Solutions relying on calculator technology are not acceptable in this question.

(i)

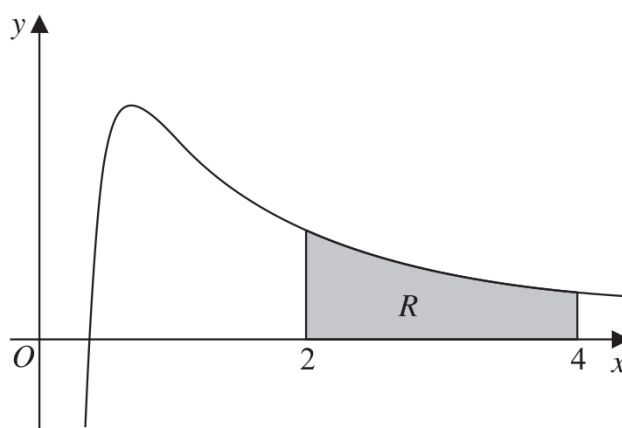


Figure 2

Figure 2 shows a sketch of part of a curve with equation

$$y = \frac{8\sqrt{x} - 5}{2x^2} \quad x > 0$$

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The region R , shown shaded in Figure 2, is bounded by the curve, the line with equation $x = 2$, the x -axis and the line with equation $x = 4$

Find the exact area of R .

(5)

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8. Solutions relying on calculator technology are not acceptable in this question.

(ii) Find the value of the constant k such that

$$\int_{-3}^6 \left(\frac{1}{2}x^2 + k \right) dx = 55$$

(4)

Adaptive unit 2 · WMA12-01_2019_Oct_Q08

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4.

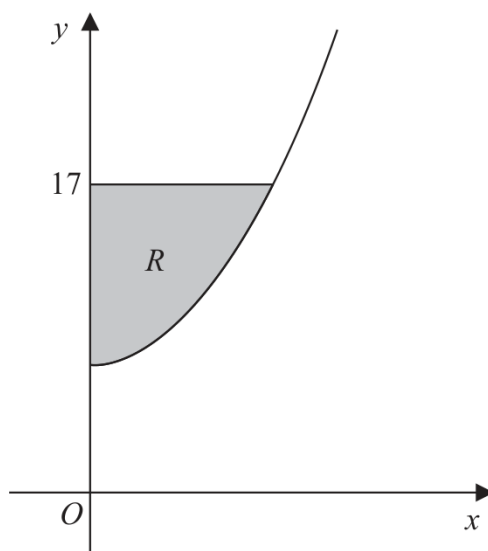
**Figure 1**

Figure 1 shows a sketch of the curve with equation

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$$y = 2x^2 + 7$$
$$x \geq 0$$

The finite region R , shown shaded in Figure 1, is bounded by the curve, the y -axis and the line with equation $y = 17$

Find the exact area of R .

(6)

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9.

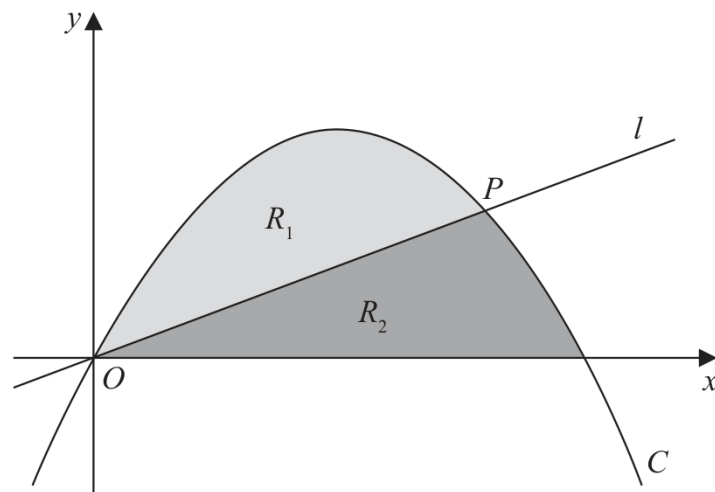
**Figure 2**

Figure 2 shows

- the curve C with equation $y = x - x^2$
- the line l with equation $y = mx$, where m is a constant and $0 < m < 1$

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The line and the curve intersect at the origin O and at the point P .

(a) Find, in terms of m , the coordinates of P .

(2)

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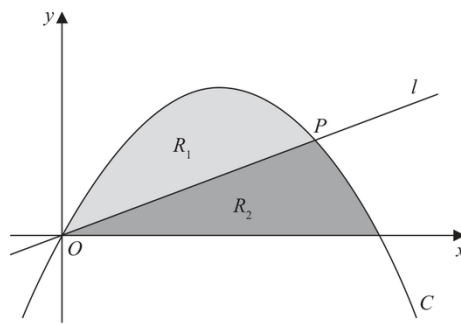


Figure 2

Figure 2 shows

- the curve C with equation $y = x - x^2$
- the line l with equation $y = mx$, where m is a constant and $0 < m < 1$

The line and the curve intersect at the origin O and at the point P .The region R_1 , shown shaded in Figure 2, is bounded by C and l .(b) Show that the area of R_1 is

$$\frac{(1-m)^3}{6}$$

(5)

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9.

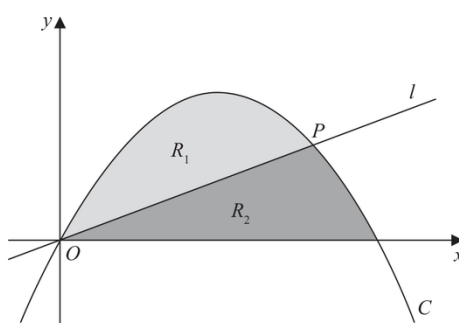


Figure 2

Figure 2 shows

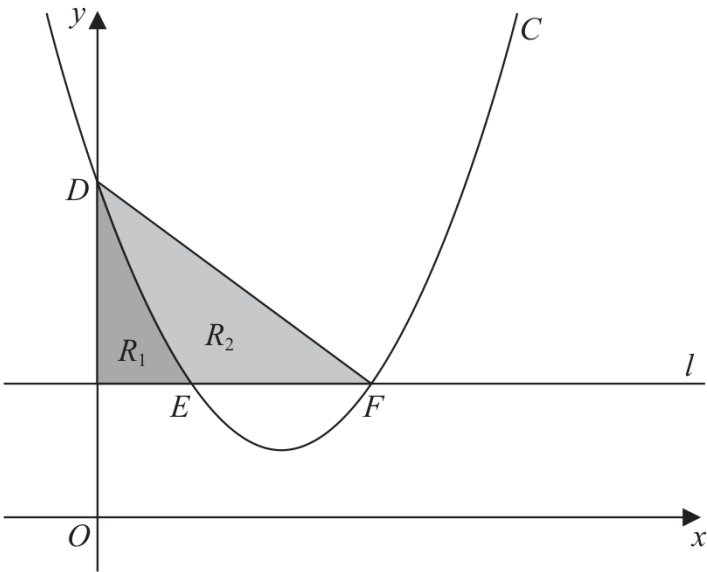
- the curve C with equation $y = x - x^2$
- the line l with equation $y = mx$, where m is a constant and $0 < m < 1$

The line and the curve intersect at the origin O and at the point P .The region R_2 , also shown shaded in Figure 2, is bounded by C , the x -axis and l .Given that the area of R_1 is equal to the area of R_2 (c) find the exact value of m .

(3)

Adaptive unit 3 · WMA12-01_2022_Jan_Q09

9. In this question you must show all stages of your working.
Solutions based entirely on calculator technology are not acceptable.



Adaptive unit 1 · WMA12-01_2023_Jan_Q09

Figure 3

Figure 3 shows

- the curve C with equation $y = x^2 - 4x + 5$
- the line l with equation $y = 2$

The curve C intersects the y -axis at the point D .

(a) Write down the coordinates of D .

(1)

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9.

In this question you must show all stages of your working.

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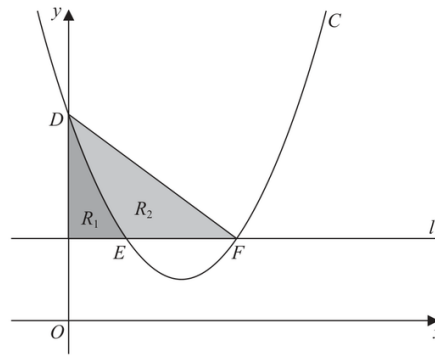


Figure 3

Figure 3 shows

- the curve C with equation $y = x^2 - 4x + 5$
- the line l with equation $y = 2$

The curve C intersects the y -axis at the point D .

The curve C intersects the line l at the points E and F , as shown in Figure 3.

(b) Find the x coordinate of E and the x coordinate of F .

(2)

Adaptive unit 2 · WMA12-01_2023_Jan_Q09

9.

In this question you must show all stages of your working.

Solutions based entirely on calculator technology are not acceptable.

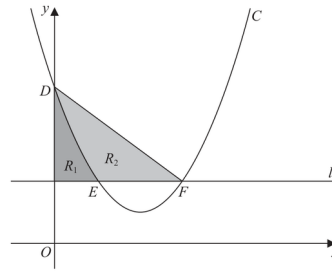


Figure 3

Figure 3 shows

- the curve C with equation $y = x^2 - 4x + 5$
- the line l with equation $y = 2$

The curve C intersects the y -axis at the point D .

Shown shaded in Figure 3 is

- the region R_1 which is bounded by C , l and the y -axis
- the region R_2 which is bounded by C and the line segments EF and DF

Given that $\frac{\text{area of } R_1}{\text{area of } R_2} = k$, where k is a constant,

- (c) use algebraic integration to find the exact value of k , giving your answer as a simplified fraction.

(5)

Adaptive unit 3 · WMA12-01_2023_Jan_Q09

10. The curve C has equation

$$y = \frac{(x - k)^2}{\sqrt{x}} \quad x > 0$$

where k is a **positive** constant.

(a) Show that

$$\int_1^{16} \frac{(x - k)^2}{\sqrt{x}} \, dx = ak^2 + bk + \frac{2046}{5}$$

where a and b are integers to be found.

(5)

Adaptive unit 1 · WMA12-01_2023_MayJune_Q10

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10. The curve C has equation

$$y = \frac{(x-k)^2}{\sqrt{x}} \quad x > 0$$

where k is a **positive** constant.

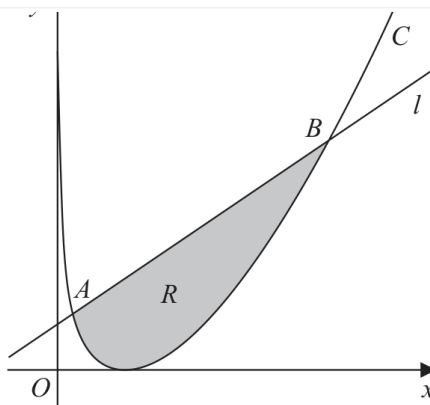


Figure 1

Figure 1 shows a sketch of the curve C and the line l .

Given that l intersects C at the point $A(1, 9)$ and at the point $B(16, q)$ where q is a constant,

(b) show that $k = 4$

(2)

Adaptive unit 2 · WMA12-01_2023_MayJune_Q10

10. The curve C has equation

$$y = \frac{(x - k)^2}{\sqrt{x}} \quad x > 0$$

where k is a **positive** constant.

The region R , shown shaded in Figure 1, is bounded by C and l

Using the answers to parts (a) and (b),

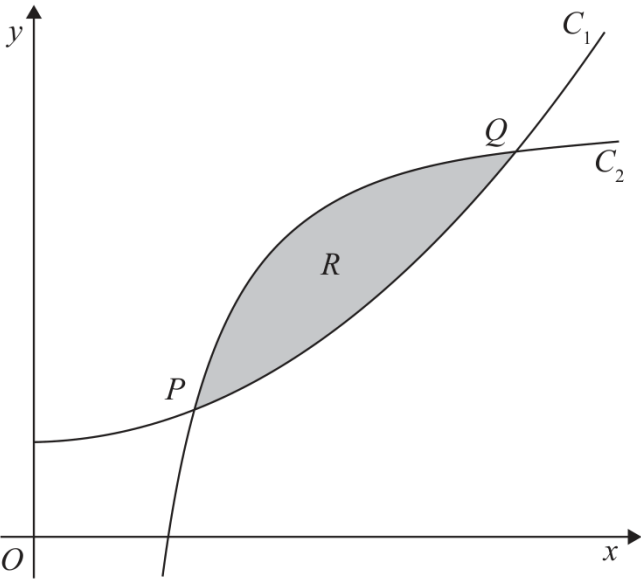
(c) find the area of region R

(3)

Adaptive unit 3 · WMA12-01_2023_MayJune_Q10

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8. In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.



Adaptive unit 1 · WMA12-01_2024_Oct_Q08

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Figure 1

Figure 1 shows a sketch of part of the curve C_1 with equation

$$y = x^2 + 3 \quad x > 0$$

and part of the curve C_2 with equation

$$y = 13 - \frac{9}{x^2} \quad x > 0$$

The curves C_1 and C_2 intersect at the points P and Q as shown in Figure 1.

(a) Use algebra to find the x coordinate of P and the x coordinate of Q .

(4)

Adaptive unit 1 · WMA12-01_2024_Oct_Q08

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8.

In this question you must show all stages of your working.

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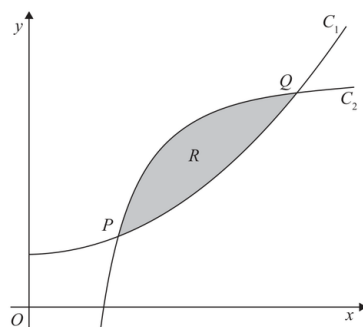


Figure 1

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$$y = x^2 + 3 \quad x > 0$$

and part of the curve C_2 with equation

$$y = 13 - \frac{9}{x^2} \quad x > 0$$

The curves C_1 and C_2 intersect at the points P and Q as shown in Figure 1.The finite region R , shown shaded in Figure 1, is bounded by C_1 and C_2 (b) Use algebraic integration to find the exact area of R .

(4)

Adaptive unit 2 · WMA12-01_2024_Oct_Q08

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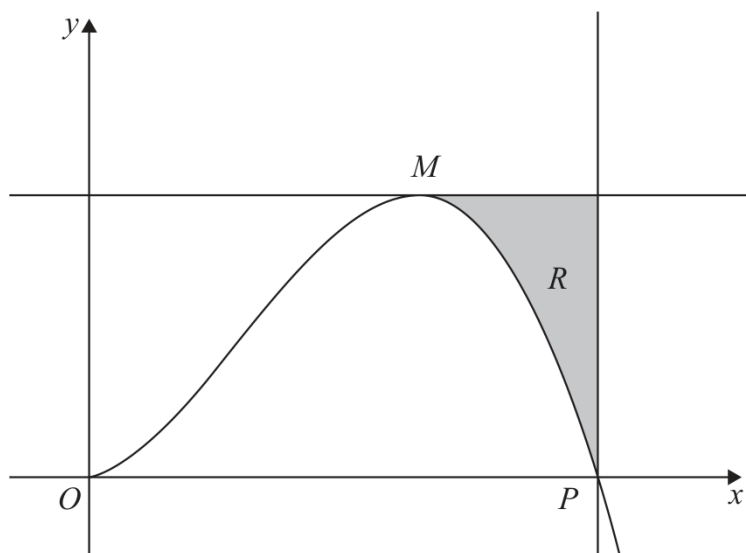


Figure 3

**In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.**

Adaptive unit 1 · WMA12-01_2025_Jan_Q09

Figure 3 shows a sketch of part of the curve with equation

$$y = \frac{9x^2(5 - \sqrt{x})}{5} \quad x \geq 0$$

The curve has a turning point at the point M , as shown in Figure 3.

(a) Using calculus, find the coordinates of M .

(5)

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9.

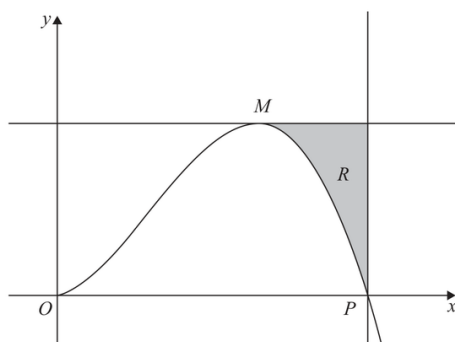


Figure 3

In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.

Figure 3 shows a sketch of part of the curve with equation

$$y = \frac{9x^2(5 - \sqrt{x})}{5} \quad x \geq 0$$

The curve has a turning point at the point M , as shown in Figure 3.

The curve crosses the x -axis at the point P , as shown in Figure 3.

(b) Use algebra to find the x coordinate of P .

(2)

Adaptive unit 2 · WMA12-01_2025_Jan_Q09

9.

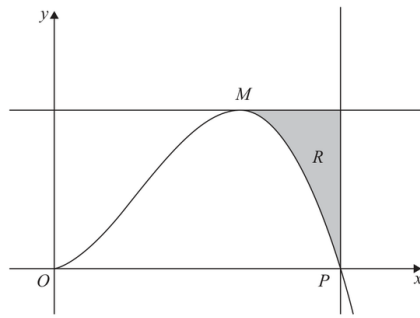


Figure 3

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Figure 3 shows a sketch of part of the curve with equation

$$y = \frac{9x^2(5 - \sqrt{x})}{5} \quad x \geq 0$$

The curve has a turning point at the point M , as shown in Figure 3.

The finite region R , shown shaded in Figure 3, is bounded by the curve, the line through M parallel to the x -axis and the line through P parallel to the y -axis.

- (c) Use algebraic integration to find the area of R , giving your answer to one decimal place.

(5)

Adaptive unit 3 · WMA12-01_2025_Jan_Q09

3:

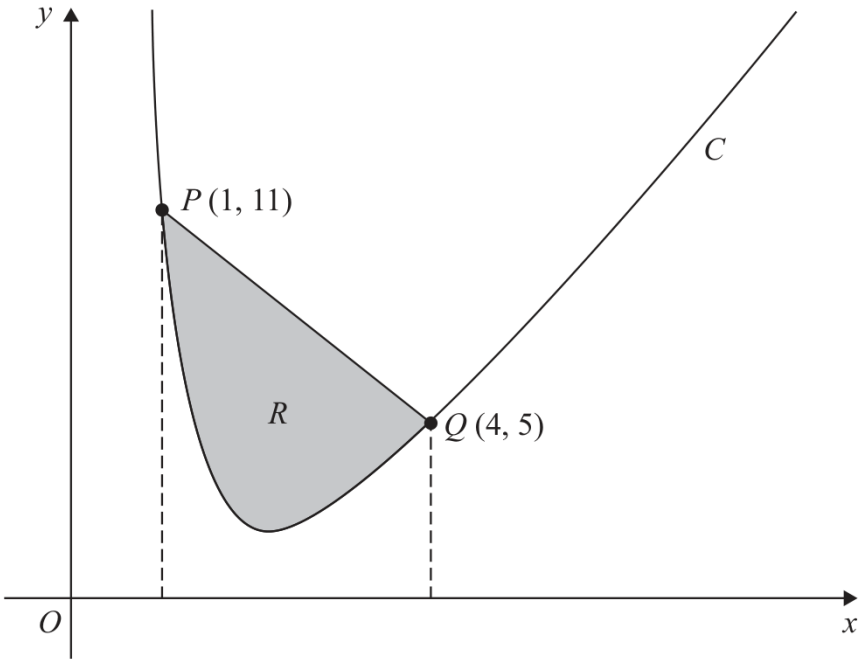


Figure 1

Adaptive unit 1 · WMA12-01_2025_MayJune_Q03

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(5)

7.

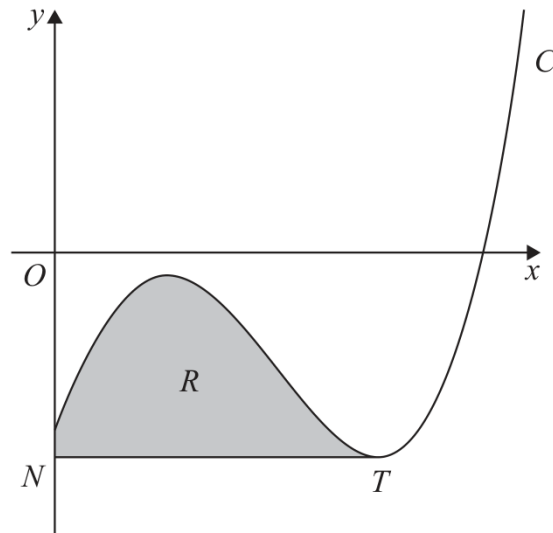


Figure 3

In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.

Adaptive unit 1 · WMA12-01_2025_Oct_Q07

Figure 3 shows a sketch of part of the curve C with equation

$$y = x^3 - 4x^{\frac{5}{2}} - kx^{\frac{1}{2}} + 28x - 44 \quad x \geq 0$$

where k is a positive constant.

(a) Find $\frac{dy}{dx}$ in simplest form.

(2)

Adaptive unit 1 · WMA12-01_2025_Oct_Q07

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7.

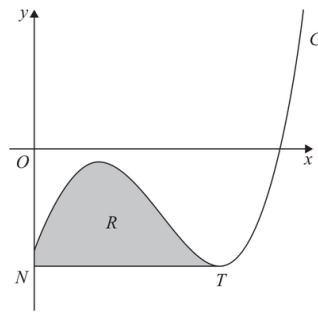


Figure 3

In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.

Figure 3 shows a sketch of part of the curve C with equation

$$y = x^3 - 4x^{\frac{5}{2}} - kx^{\frac{1}{2}} + 28x - 44 \quad x \geq 0$$

where k is a positive constant.

The point T , shown in Figure 3, is a minimum stationary point on C .

Given that the x coordinate of T is 9

(b) show that $k = 6$

(2)

Adaptive unit 2 · WMA12-01_2025_Oct_Q07

7.

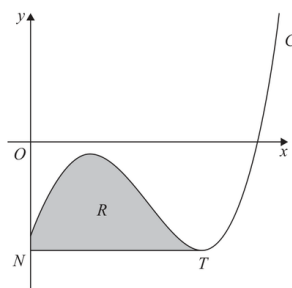


Figure 3

In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.

Figure 3 shows a sketch of part of the curve C with equation

$$y = x^3 - 4x^{\frac{5}{2}} - kx^{\frac{1}{2}} + 28x - 44 \quad x \geq 0$$

where k is a positive constant.

The line through T parallel to the x -axis meets the y -axis at the point N .

The finite region R , shown shaded in Figure 3, is bounded by C , the y -axis and the line segment NT .

- (c) Use algebraic integration to find the area of R , giving the answer to 3 significant figures.

(6)

Adaptive unit 3 · WMA12-01_2025_Oct_Q07

TOPIC 12 | FAMILY 02

Power Integration

WHAT THIS FAMILY ASKS

Recognise the power integration demand and complete the required reasoning in a clear, exam-ready order.

VARIANT ROUTE

V01 core family move

2 classified questions

12 total marks

7.

$$f(x) = Ax^3 + 6x^2 - 4x + B$$

where A and B are constants.

Given that

- $(x + 2)$ is a factor of $f(x)$

- $\int_3^5 f(x) dx = 176$

find the value of A and the value of B .

(7)

Adaptive unit 1 · WMA12-01_2022_MayJune_Q07

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4. **In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.**

Given that k is a positive constant and

$$\int_k^{2k} \left(\frac{12}{x^2} + 4 \right) dx = 14$$

find the possible values of k .

(5)

Adaptive unit 1 · WMA12-01_2026_Jan_Q04

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TOPIC 12 | FAMILY 03

Transformed Estimates

WHAT THIS FAMILY ASKS

Recognise the transformed estimates demand and complete the required reasoning in a clear, exam-ready order.

VARIANT ROUTE

V01 For $f(x+c)$ or $f(-x)$, check the limits by substitution.

V02 Do not round the known estimate too early.

8 classified questions

53 total marks

1.

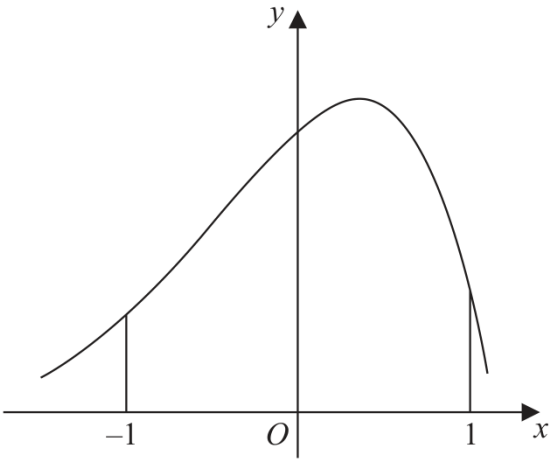


Figure 1

Figure 1 shows a sketch of part of the curve with equation $y = f(x)$

The table below shows some corresponding values of x and y for this curve.

The values of y are given to 3 decimal places.

Adaptive unit 1 · WMA12-01_2023_Jan_Q01

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x	-1	-0.5	0	0.5	1
y	2.287	4.470	6.719	7.291	2.834

Using the trapezium rule with all the values of y in the given table,

(a) obtain an estimate for

$$\int_{-1}^1 f(x) \, dx$$

giving your answer to 2 decimal places.

(3)

Adaptive unit 1 · WMA12-01_2023_Jan_Q01

1.

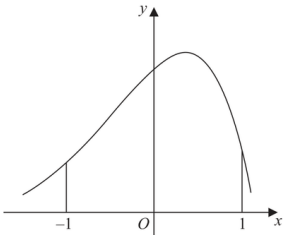


Figure 1

Figure 1 shows a sketch of part of the curve with equation $y = f(x)$
The table below shows some corresponding values of x and y for this curve.
The values of y are given to 3 decimal places.

x	-1	-0.5	0	0.5	1
y	2.287	4.470	6.719	7.291	2.834

Using the trapezium rule with all the values of y in the given table,

(b) Use your answer to part (a) to estimate

(i) $\int_{-1}^1 (f(x) - 2) \, dx$

(ii) $\int_1^3 f(x - 2) \, dx$

(3)

4. (a) Sketch the curve with equation

$$y = a^{-x} + 4$$

where a is a constant and $a > 1$

On your sketch show

- the coordinates of the point of intersection of the curve with the y -axis
- the equation of the asymptote to the curve.

(3)

Adaptive unit 1 · WMA12-01_2024_Jan_Q04

DR ESAM AHMED

x	-4	-1.5	1	3.5	6	8.5
y	13	6.280	4.577	4.146	4.037	4.009

The table above shows corresponding values of x and y for $y = 3^{-\frac{1}{2}x} + 4$

The values of y are given to four significant figures, as appropriate.

Using the trapezium rule with all the values of y in the table,

(b) find an approximate value for

$$\int_{-4}^{8.5} \left(3^{-\frac{1}{2}x} + 4 \right) dx$$

giving your answer to two significant figures.

(3)

(c) Using the answer to part (b), find an approximate value for

(i) $\int_{-4}^{8.5} \left(3^{-\frac{1}{2}x} \right) dx$

(ii) $\int_{-4}^{8.5} \left(3^{-\frac{1}{2}x} + 4 \right) dx + \int_{-8.5}^4 \left(3^{\frac{1}{2}x} + 4 \right) dx$

(3)

Adaptive unit 3 · WMA12-01_2024_Jan_Q04

DR ESAM AHMED

2. The table shows corresponding values of x and y for a continuous curve with equation $y = f(x)$ between $x = -4$ and $x = 5$, where a is a constant.

x	-4	-2.5	-1	0.5	2	3.5	5
y	4.16	2.91	a	1.73	1.37	1.43	2.28

The trapezium rule is used with all the y values in the table to find an approximation for

$$\int_{-4}^5 f(x) \, dx$$

Given that the value of this approximation is 19.3

- (a) find the value of the constant a to 3 significant figures.

(3)

2. The table shows corresponding values of x and y for a continuous curve with equation $y = f(x)$ between $x = -4$ and $x = 5$, where a is a constant.

x	-4	-2.5	-1	0.5	2	3.5	5
y	4.16	2.91	a	1.73	1.37	1.43	2.28

The trapezium rule is used with all the y values in the table to find an approximation for

$$\int_{-4}^5 f(x) \, dx$$

Given that the value of this approximation is 19.3

- (b) Use the given answer of 19.3 to find an approximate value for

$$\int_{-4}^5 (2f(x) - 3) \, dx$$

(2)

1. A continuous curve has equation $y = f(x)$.

A table of values of x and y for $y = f(x)$ is shown below.

x	0.5	1.75	3	4.25	5.5
y	3.479	6.101	7.448	6.823	5.182

Using the trapezium rule with all the values of y in the given table,

(a) find an estimate for

$$\int_{0.5}^{5.5} f(x) \, dx$$

giving your answer to one decimal place.

(3)

Adaptive unit 1 · WMA12-01_2024_Oct_Q01

DR ESAM AHMED

1. A continuous curve has equation $y = f(x)$.
A table of values of x and y for $y = f(x)$ is shown below.

x	0.5	1.75	3	4.25	5.5
y	3.479	6.101	7.448	6.823	5.182

Using the trapezium rule with all the values of y in the given table,

(b) Using your answer to part (a) and making your method clear, estimate

$$\int_{0.5}^{5.5} (f(x) + 4x) \, dx$$

(2)

DR ESAM AHMED

6. (a) Sketch the curve with equation

$$y = a^x + 4$$

where a is a positive constant greater than 1

On your sketch, show

- the coordinates of the point of intersection of the curve with the y -axis
- the equation of the asymptote of the curve

(3)

Adaptive unit 1 · WMA12-01_2024_MayJune_Q06

DR ESAM AHMED

x	2	2.3	2.6	2.9	3.2	3.5
y	0	0.3246	0.8629	1.6643	2.7896	4.3137

The table shows corresponding values of x and y for

$$y = 2^x - 2x$$

with the values of y given to 4 decimal places as appropriate.

Using the trapezium rule with all the values of y in the given table,

(b) obtain an estimate for $\int_2^{3.5} (2^x - 2x) \, dx$, giving your answer to 2 decimal places.

(3)

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(c) Using your answer to part (b) and making your method clear, estimate

(i) $\int_2^{3.5} (2^x + 2x) \, dx$

(ii) $\int_2^{3.5} (2^{x+1} - 4x) \, dx$

(3)

Adaptive unit 3 · WMA12-01_2024_MayJune_Q06

DR ESAM AHMED

5:

x	-2	-0.5	1	2.5	4	5.5	7
y	12	4.243	1.5	0.530	0.188	0.066	0.023

The table above shows corresponding values of x and y for

$$y = 3\left(\frac{1}{2}\right)^x$$

The values of y are given to 3 decimal places as appropriate.

- (a) Using the trapezium rule with all the values of y in the given table, obtain an estimate for

$$\int_{-2}^7 3\left(\frac{1}{2}\right)^x \, dx$$

giving the answer to one decimal place.

(3)

Adaptive unit 1 · WMA12-01_2025_MayJune_Q05

DR ESAM AHMED

5:

<i>x</i>	−2	−0.5	1	2.5	4	5.5	7
<i>y</i>	12	4.243	1.5	0.530	0.188	0.066	0.023

The table above shows corresponding values of *x* and *y* for

$$y = 3\left(\frac{1}{2}\right)^x$$

The values of *y* are given to 3 decimal places as appropriate.

Using the answer to part (a) and making your method clear, estimate

(b) (i) $\int_{-2}^7 3\left(\frac{1}{2}\right)^{x+2} \mathrm{d}x$

(ii) $\int_{-2}^7 \left(2^{-x} + 2x\right) \mathrm{d}x$

(3)

3.

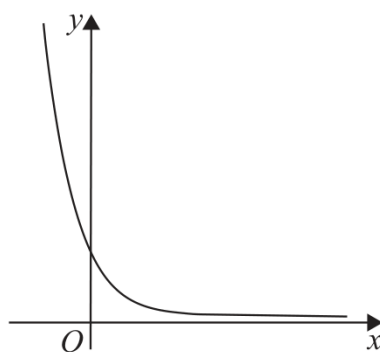


Figure 1

**In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.**

Figure 1 shows a sketch of the curve with equation $y = 3 \times 2^{-x}$

The point $P(k, 300\,000)$ lies on the curve.

Adaptive unit 1 · WMA12-01_2025_Oct_Q03

(a) Use logarithms to find the value of k to 2 decimal places.

(2)

Adaptive unit 1 · WMA12-01_2025_Oct_Q03

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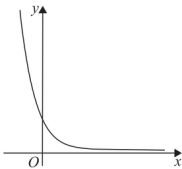


Figure 1

In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.

Figure 1 shows a sketch of the curve with equation $y = 3 \times 2^{-x}$

The point $P(k, 300\,000)$ lies on the curve.

x	-0.5	1	2.5	4.0	5.5	7
y	4.243	1.5	0.530	0.188	0.066	0.023

The table shows corresponding values of x and y for $y = 3 \times 2^{-x}$

The values of y are given to 3 decimal places where appropriate.

- (b) (i) Use the trapezium rule, with all the values of y from the table, to find an approximate value, to 2 decimal places, for

$$\int_{-0.5}^7 3 \times 2^{-x} \, dx$$

(3)

(ii) Use your answer to part (b)(i) to estimate

$$\int_{-0.5}^7 2^{-x} \, dx + \int_{-7}^{0.5} 2^x \, dx$$

(2)

Adaptive unit 2 · WMA12-01_2025_Oct_Q03

DR ESAM AHMED

2.

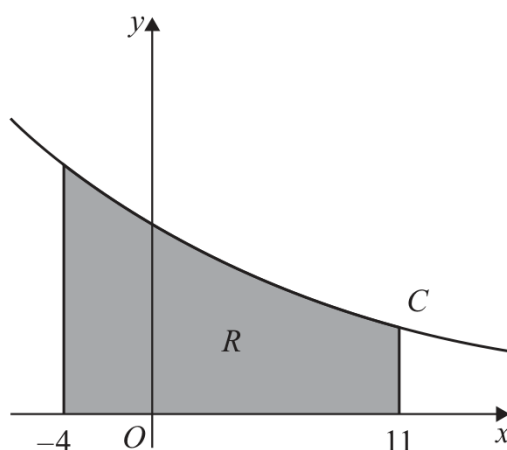
**Figure 1**

Figure 1 shows a sketch of part of the curve C with equation $y = 2^{-0.1x}$

The table below shows corresponding values of x and y for $y = 2^{-0.1x}$

The y values are given to 4 decimal places.

Adaptive unit 1 · WMA12-01_2026_Jan_Q02

x	-4	-1	2	5	8	11
y	1.3195	1.0718	0.8706	0.7071	0.5743	0.4665

The region R , shown shaded in Figure 1, is bounded by C , the x -axis and the lines with equations $x = -4$ and $x = 11$

- (a) Use the trapezium rule with all the values of y in the table to find an estimate for the area of R . Give the answer to 2 decimal places.

(3)

DR ESAM AHMED

2.

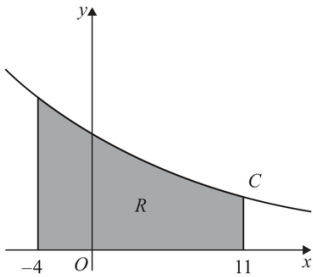


Figure 1

Figure 1 shows a sketch of part of the curve C with equation $y = 2^{-0.1x}$
The table below shows corresponding values of x and y for $y = 2^{-0.1x}$
The y values are given to 4 decimal places.

x	-4	-1	2	5	8	11
y	1.3195	1.0718	0.8706	0.7071	0.5743	0.4665

The region R , shown shaded in Figure 1, is bounded by C , the x -axis and the lines with equations $x = -4$ and $x = 11$

(b) State how you would use the trapezium rule to get a more accurate estimate for the true area of R .

(1)

2.

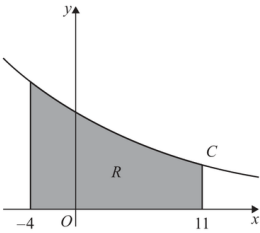


Figure 1

Figure 1 shows a sketch of part of the curve C with equation $y = 2^{-0.1x}$
The table below shows corresponding values of x and y for $y = 2^{-0.1x}$
The y values are given to 4 decimal places.

x	-4	-1	2	5	8	11
y	1.3195	1.0718	0.8706	0.7071	0.5743	0.4665

The region R , shown shaded in Figure 1, is bounded by C , the x -axis and the lines with equations $x = -4$ and $x = 11$

Using the answer to part (a) and showing your working,
(c) estimate the value of

$$\int_{-4}^{11} 2^{3-0.1x} \, dx$$

(2)

TOPIC 12 | FAMILY 04

Numerical Integration

WHAT THIS FAMILY ASKS

Recognise the numerical integration demand and complete the required reasoning in a clear, exam-ready order.

VARIANT ROUTE

V01 core family move

10 classified questions

69 total marks

5. (a) Given $0 < a < 1$, sketch the curve with equation

$$y = a^x$$

showing the coordinates of the point at which the curve crosses the y-axis.

(2)

Adaptive unit 1 · WMA12-01_2019_Oct_Q05

DR ESAM AHMED

x	2	2.5	3	3.5	4
y	4.25	6.427	9.125	12.34	16.06

The table above shows corresponding values of x and y for $y = x^2 + \left(\frac{1}{2}\right)^x$

The values of y are given to 4 significant figures as appropriate.

Using the trapezium rule with all the values of y in the given table,

(b) obtain an estimate for $\int_2^4 \left(x^2 + \left(\frac{1}{2}\right)^x\right) dx$

(3)

DR ESAM AHMED

Using your answer to part (b) and making your method clear, estimate

(c) $\int_2^4 \left(x(x - 3) + \left(\frac{1}{2} \right)^x \right) dx$

(2)

Adaptive unit 3 · WMA12-01_2019_Oct_Q05

DR ESAM AHMED

1. The table below shows corresponding values of x and y for $y = \log_2(2x)$

The values of y are given to 2 decimal places as appropriate.

x	2	5	8	11	14
y	2	3.32	4	4.46	4.81

Using the trapezium rule with all the values of y in the given table,

(a) obtain an estimate for $\int_2^{14} \log_2(2x)dx$, giving your answer to one decimal place. (3)

Adaptive unit 1 · WMA12-01_2020_Jan_Q01

DR ESAM AHMED

1. The table below shows corresponding values of x and y for $y = \log_2(2x)$

The values of y are given to 2 decimal places as appropriate.

x	2	5	8	11	14
y	2	3.32	4	4.46	4.81

Using the trapezium rule with all the values of y in the given table,

14

Using your answer to part (a) and making your method clear, estimate

(b) (i) $\int_2^{14} \frac{\log_2(4x^2)}{5} dx$

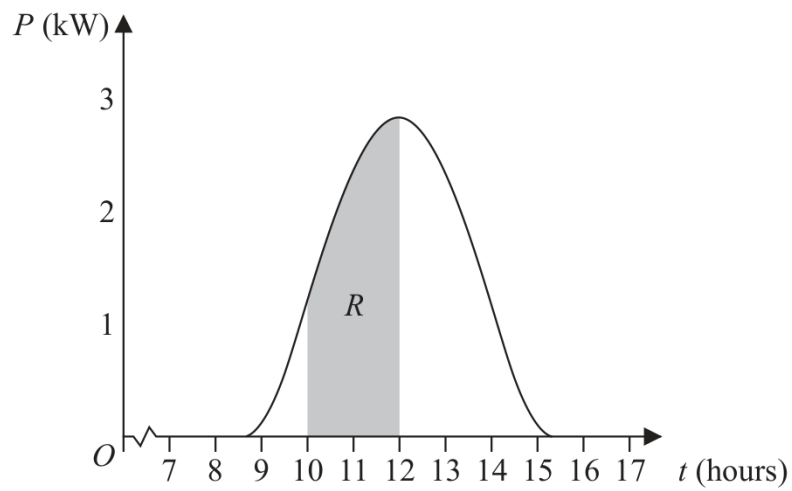
(ii) $\int_2^{14} \log_2\left(\frac{2}{x}\right) dx$

(4)

Adaptive unit 2 · WMA12-01_2020_Jan_Q01

DR ESAM AHMED

7.

**Figure 1**

Solar panels are installed on the roof of a building.

The power, P , produced on a particular day, in kW, can be modelled by the equation

Adaptive unit 1 · WMA12-01_2021_Jan_Q07

$$8.5 \leq t \leq 15.2$$

A table of values of t and P is shown below, with the values of P given to 4 significant figures where appropriate.

Time, t (hours)	10	10.5	11	11.5	12
Power, P (kW)		1.882	2.45		2.95

- (a) Use the given equation to complete the table, giving the values of P to 4 significant figures where appropriate.

(2)

Adaptive unit 1 · WMA12-01_2021_Jan_Q07

7.

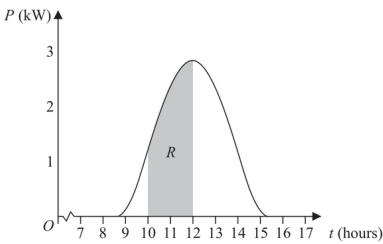


Figure 1

Solar panels are installed on the roof of a building.

The power, P , produced on a particular day, in kW, can be modelled by the equation

$$P = 0.95 + 2^{t-12} + 2^{12-t} - (t - 12)^2 \quad 8.5 \leq t \leq 15.2$$

where t is the time in hours after midnight. The graph of P against t is shown in Figure 1.

A table of values of t and P is shown below, with the values of P given to 4 significant figures where appropriate.

Time, t (hours)	10	10.5	11	11.5	12
Power, P (kW)		1.882	2.45		2.95

- The amount of energy, in kWh, produced between 10:00 and 12:00 can be found by calculating the area of region R , shown shaded in Figure 1.
- (b) Use the trapezium rule, with all the values of P in the completed table, to find an estimate for the amount of energy produced between 10:00 and 12:00. Give your answer to 2 decimal places.
- (4)

5.

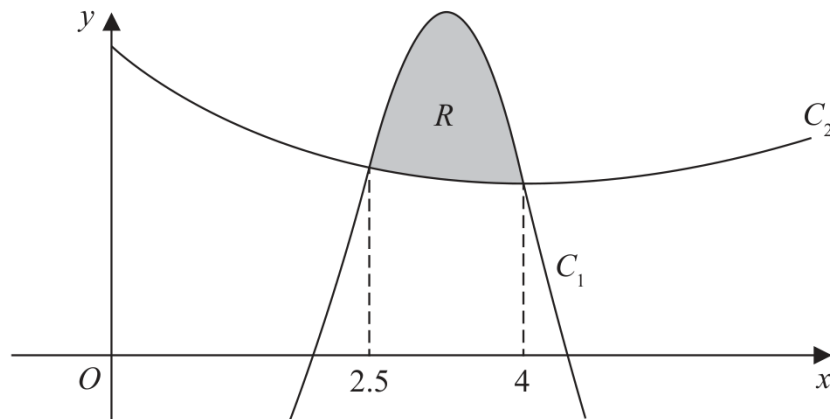
**Figure 2**

Figure 2 shows a sketch of part of the graph of the curves C_1 and C_2

The curves intersect when $x = 2.5$ and when $x = 4$

A table of values for some points on the curve C_1 is shown below, with y values given to 3 decimal places as appropriate.

Adaptive unit 1 · WMA12-01_2021_MayJune_Q05

x	2.5	2.75	3	3.25	3.5	3.75	4
y	5.453	7.764	9.375	9.964	9.367	7.626	5

Using the trapezium rule with all the values of y in the table,

- (a) find, to 2 decimal places, an estimate for the area bounded by the curve C_1 , the line with equation $x = 2.5$, the x -axis and the line with equation $x = 4$

(4)

DR ESAM AHMED

5.

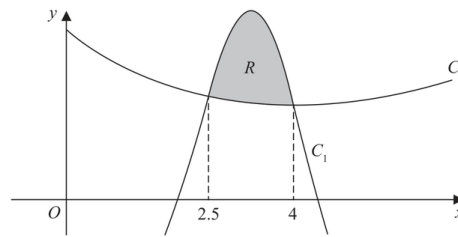


Figure 2

Figure 2 shows a sketch of part of the graph of the curves C_1 and C_2

The curves intersect when $x = 2.5$ and when $x = 4$

A table of values for some points on the curve C_1 is shown below, with y values given to 3 decimal places as appropriate.

x	2.5	2.75	3	3.25	3.5	3.75	4
y	5.453	7.764	9.375	9.964	9.367	7.626	5

Using the trapezium rule with all the values of y in the table,

The curve C_2 has equation

$$y = x^{\frac{3}{2}} - 3x + 9 \quad x > 0$$

(b) Find $\int \left(x^{\frac{3}{2}} - 3x + 9 \right) dx$

(3)

5.

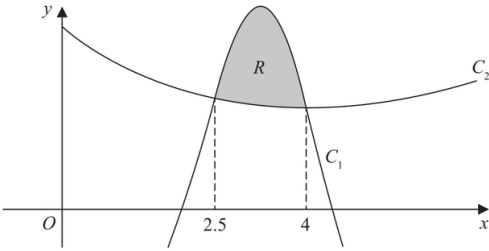


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Figure 2 shows a sketch of part of the graph of the curves C_1 and C_2

The curves intersect when $x = 2.5$ and when $x = 4$

A table of values for some points on the curve C_1 is shown below, with y values given to 3 decimal places as appropriate.

x	2.5	2.75	3	3.25	3.5	3.75	4
y	5.453	7.764	9.375	9.964	9.367	7.626	5

Using the trapezium rule with all the values of y in the table,

The region R , shown shaded in Figure 2, is bounded by the curves C_1 and C_2

- (c) Use the answers to part (a) and part (b) to find, to one decimal place, an estimate for the area of the region R .

(3)

3.

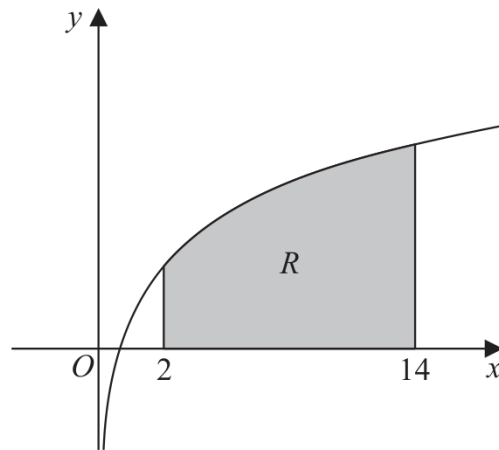
**Figure 1**

Figure 1 shows a sketch of part of the curve with equation $y = \log_{10} x$

The region R , shown shaded in Figure 1, is bounded by the curve, the line with equation $x = 2$, the x -axis and the line with equation $x = 14$

Adaptive unit 1 · WMA12-01_2021_Oct_Q03

Using the trapezium rule with four strips of equal width,
(a) show that the area of R is approximately 10.10

(3)

Adaptive unit 1 · WMA12-01_2021_Oct_Q03

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3.

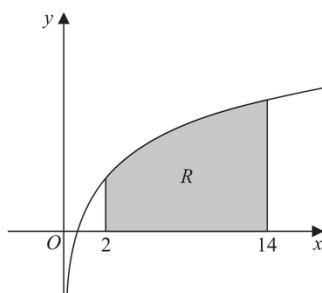


Figure 1

Figure 1 shows a sketch of part of the curve with equation $y = \log_{10} x$

The region R , shown shaded in Figure 1, is bounded by the curve, the line with equation $x = 2$, the x -axis and the line with equation $x = 14$

Using the trapezium rule with four strips of equal width,

- (b) Explain how the trapezium rule could be used to obtain a more accurate estimate for the area of R .

(1)

Adaptive unit 2 · WMA12-01_2021_Oct_Q03

3.

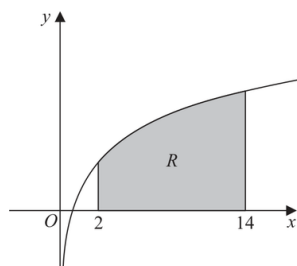


Figure 1

Figure 1 shows a sketch of part of the curve with equation $y = \log_{10} x$

The region R , shown shaded in Figure 1, is bounded by the curve, the line with equation $x = 2$, the x -axis and the line with equation $x = 14$

Using the trapezium rule with four strips of equal width,

(c) Using the answer to part (a) and making your method clear, estimate the value of

(i) $\int_2^{14} \log_{10} \sqrt{x} \, dx$

(ii) $\int_2^{14} \log_{10} 100x^3 \, dx$

(4)

Adaptive unit 3 · WMA12-01_2021_Oct_Q03

1. The table below shows corresponding values of x and y for

$$y = 2^{5-\sqrt{x}}$$

The values of y are given to 3 decimal places.

x	5	5.5	6	6.5	7
y	6.792	6.298	5.858	5.466	5.113

Using the trapezium rule with all the values of y in the given table,

- (a) obtain an estimate for

$$\int_5^7 2^{5-\sqrt{x}} dx$$

giving your answer to 2 decimal places.

(3)

Adaptive unit 1 · WMA12-01_2022_Jan_Q01

DR ESAM AHMED

1. The table below shows corresponding values of x and y for

$$y = 2^{5-\sqrt{x}}$$

The values of y are given to 3 decimal places.

x	5	5.5	6	6.5	7
y	6.792	6.298	5.858	5.466	5.113

Using the trapezium rule with all the values of y in the given table,

- (b) Using your answer to part (a) and making your method clear, estimate

(i) $\int_5^7 2^{6-\sqrt{x}} dx$

(ii) $\int_5^7 (3 + 2^{5-\sqrt{x}}) dx$

(4)

Adaptive unit 2 · WMA12-01_2022_Jan_Q01

DR ESAM AHMED

2.

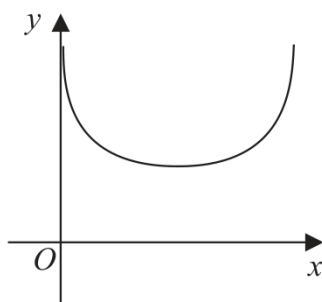
**Figure 1**

Figure 1 shows the graph of

$$y = 1 - \log_{10}(\sin x) \quad 0 < x < \pi$$

where x is in radians.

The table below shows some values of x and y for this graph, with values of y given to 3 decimal places.

Adaptive unit 1 · WMA12-01_2022_MayJune_Q02

x	0.5	1	1.5	2	2.5	3
y	1.319		1.001		1.223	1.850

(a) Complete the table above, giving values of y to 3 decimal places.

(2)

Adaptive unit 1 · WMA12-01_2022_MayJune_Q02

2.

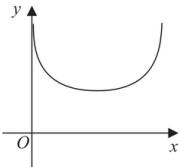


Figure 1

Figure 1 shows the graph of

$y = 1 - \log_{10}(\sin x) \qquad 0 < x < \pi$

where x is in radians.

The table below shows some values of x and y for this graph, with values of y given to 3 decimal places.

x	0.5	1	1.5	2	2.5	3
y	1.319		1.001		1.223	1.850

- (b) Use the trapezium rule with all the y values in the completed table to find, to 2 decimal places, an estimate for

$$\int_{0.5}^3 (1 - \log_{10}(\sin x)) dx$$

(3)

2.

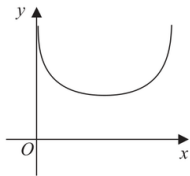


Figure 1

Figure 1 shows the graph of

$y = 1 - \log_{10}(\sin x) \qquad 0 < x < \pi$

where x is in radians.

The table below shows some values of x and y for this graph, with values of y given to 3 decimal places.

x	0.5	1	1.5	2	2.5	3
y	1.319		1.001		1.223	1.850

(c) Use your answer to part (b) to find an estimate for

$$\int_{0.5}^3 (3 + \log_{10}(\sin x)) \, dx$$

(3)

6. The curve C_1 has equation $y = f(x)$.

A table of values of x and y for $y = f(x)$ is shown below, with the y values rounded to 4 decimal places where appropriate.

x	0	0.5	1	1.5	2
y	3	2.6833	2.4	2.1466	1.92

(a) Use the trapezium rule with all the values of y in the table to find an approximation for

$$\int_0^2 f(x) \, dx$$

giving your answer to 3 decimal places.

(3)

Adaptive unit 1 · WMA12-01_2022_Oct_Q06

DR ESAM AHMED

6. The curve C_1 has equation $y = f(x)$.
A table of values of x and y for $y = f(x)$ is shown below, with the y values rounded to 4 decimal places where appropriate.

x	0	0.5	1	1.5	2
y	3	2.6833	2.4	2.1466	1.92

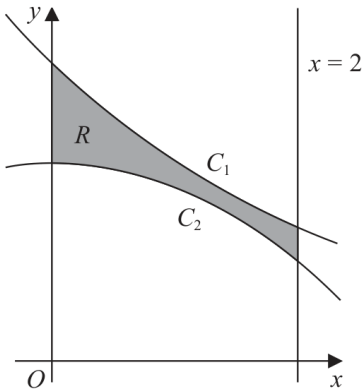


Figure 1

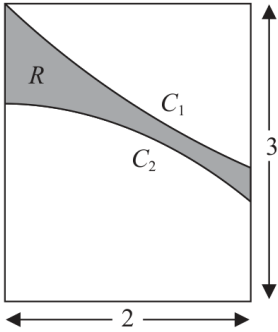


Figure 2

The region R , shown shaded in Figure 1, is bounded by

- the curve C_1
- the curve C_2 with equation $y = 2 - \frac{1}{4}x^2$

Adaptive unit 2 · WMA12-01_2022_Oct_Q06

- the line with equation $x = 2$
- the y -axis

The region R forms part of the design for a logo shown in Figure 2.

The design consists of the shaded region R inside a rectangle of width 2 and height 3

Using calculus and the answer to part (a),

(b) calculate an estimate for the percentage of the logo which is shaded.

(4)

Adaptive unit 2 · WMA12-01_2022_Oct_Q06

DR ESAM AHMED

1. The continuous curve C has equation $y = f(x)$.

A table of values of x and y for $y = f(x)$ is shown below.

x	4.0	4.2	4.4	4.6	4.8	5.0
y	9.2	8.4556	3.8512	5.0342	7.8297	8.6

Use the trapezium rule with all the values of y in the table to find an approximation for

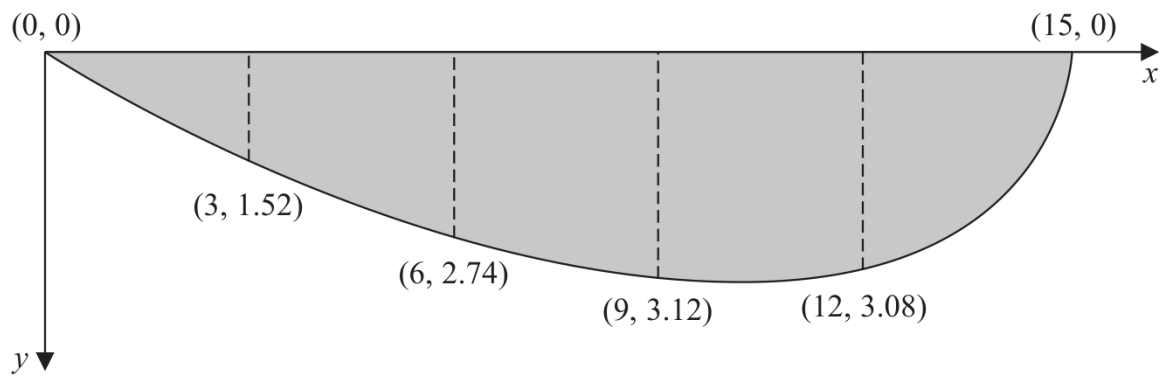
$$\int_4^5 f(x) \, dx$$

giving your answer to 3 decimal places.

(3)

DR ESAM AHMED

6.

**Figure 1**

A river is being studied.

At one particular place, the river is 15 m wide.

The depth, y metres, of the river is measured at a point x metres from one side of the river.

Figure 1 shows a plot of the cross-section of the river and the coordinate values (x, y)

Adaptive unit 1 · WMA12-01_2023_Oct_Q06

(a) Use the trapezium rule with all the y values given in Figure 1 to estimate the cross-sectional area of the river.

(3)

Adaptive unit 1 · WMA12-01_2023_Oct_Q06

DR ESAM AHMED

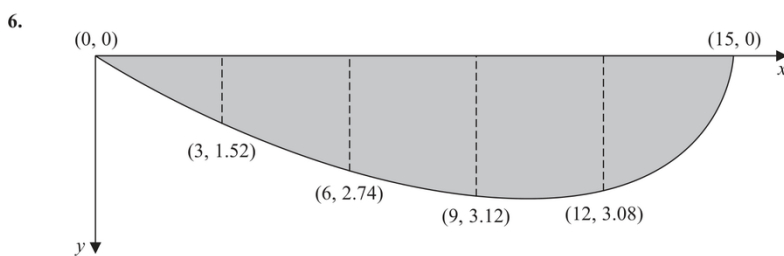


Figure 1

A river is being studied.

At one particular place, the river is 15 m wide.

The depth, y metres, of the river is measured at a point x metres from one side of the river.

Figure 1 shows a plot of the cross-section of the river and the coordinate values (x, y)

The water in the river is modelled as flowing at a constant speed of 1.5 m s^{-1} across the whole of the cross-section.

- (b) Use the model and the answer to part (a) to estimate the volume of water flowing through this section of the river each minute, giving your answer in m^3 to 2 significant figures.

(2)

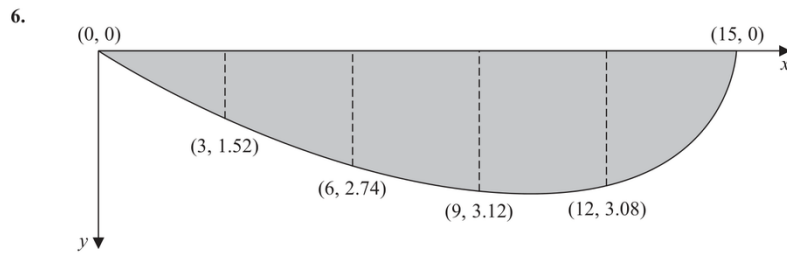


Figure 1

A river is being studied.

At one particular place, the river is 15 m wide.

The depth, y metres, of the river is measured at a point x metres from one side of the river.

Figure 1 shows a plot of the cross-section of the river and the coordinate values (x, y)

Assuming the model,

- (c) state, giving a reason for your answer, whether your answer for part (b) is an overestimate or an underestimate of the true volume of water flowing through this section of the river each minute.

(1)

Adaptive unit 3 · WMA12-01_2023_Oct_Q06